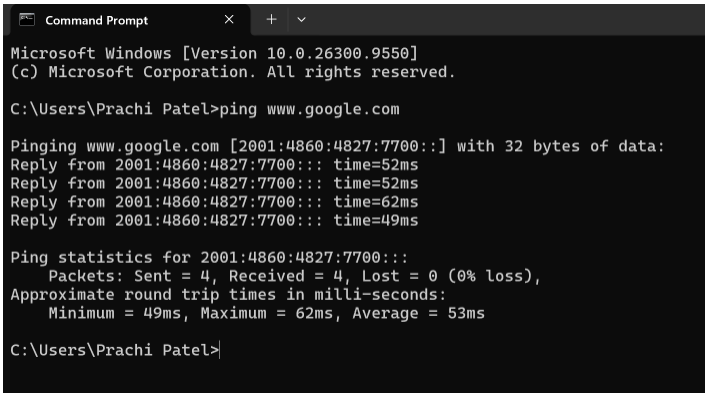
Troubleshooting Tools

1. Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.



2. Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.



Average response time: 53 ms

3. Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

```
C:\Users\Prachi Patel>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [23.212.50.126]
over a maximum of 30 hops:

  1     2 ms     3 ms     3 ms  Unit [192.168.1.1]
  2    45 ms    37 ms    42 ms  10.240.8.38
  3     *        *        *     Request timed out.
  4    29 ms    33 ms    39 ms  182.79.249.105
  5    57 ms    46 ms    69 ms  182.79.224.126
  6    54 ms    76 ms    40 ms  ae7.r02.bom01.ien.netarch.akamai.com [23.56.136.206]
  7    92 ms    45 ms     *     ae11.tata-bom3.netarch.akamai.com [104.70.118.135]
  8    60 ms    50 ms    39 ms  a23-212-50-126.deploy.static.akamaitechnologies.com [23.212.50.126]

Trace complete.

C:\Users\Prachi Patel>
```

Number of hops to reach www.flipkart.com: 8 hops.

4. Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

```
C:\Users\Prachi Patel>nslookup www.spotify.com
Server:  UnKnown
Address:  fe80::e88:32ff:fe9f:e60c

Non-authoritative answer:
Name:    atc.spotify.map.fastly.net
Addresses:  2a04:4e42:31::810
          151.101.211.42
Aliases:  www.spotify.com


C:\Users\Prachi Patel>
```

**5. A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

Hint: Think about checking both network connectivity and DNS resolution.**

Troubleshooting Sequence

If my friend cannot access Zomato's website, I would use the following three commands in order:

1. ipconfig – First, I would check the laptop's network configuration. It shows the IPv4 address, subnet mask, and default gateway. This helps confirm whether the laptop is properly connected to the local network.

2. ping www.zomato.com – Next, I would use the ping command to check whether the laptop can communicate with Zomato's server. If replies are received, basic network connectivity is working.
3. nslookup www.zomato.com – Finally, I would check DNS resolution. This command verifies whether the domain name www.zomato.com can be successfully converted into an IP address.